



NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission Template

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YEAR : III

COLLEGE CODE : 106

COLLEGE NAME : SHREE RAGHAVENDRA ARTS AND SCIENCE COLLEGE

DEPARTMENT : COMPUTER SCIENCE

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TITLE : TRAINER SESSION ATTENDANCE

DATE : 24/06/2026

GITHUB LINK :

<https://github.com/Archana25082007/Archana-NM-Internship-.git>

1. ABSTRACT

The Trainer Session Attendance Dashboard is a web-based application developed using React that helps training managers efficiently manage and monitor student attendance across multiple batches. The system allows trainers to mark attendance as Present or Absent, view attendance records batch-wise, filter attendance based on dates, and calculate attendance percentages automatically. This application simplifies attendance management, reduces manual effort, and provides accurate attendance statistics for better decision-making.

2. PROBLEM STATEMENT

Training institutes and organizations often face difficulties in maintaining attendance records manually. Managing attendance for multiple batches and generating attendance statistics becomes time-consuming and error-prone. There is a need for a digital system that enables trainers to record attendance efficiently and monitor attendance details batch-wise and date-wise.

3. OBJECTIVES

- *To provide an interface for marking student attendance.*
- *To maintain attendance records for multiple batches.*
- *To enable date-wise filtering of attendance data.*
- *To calculate attendance percentages automatically.*
- *To simplify attendance tracking and reduce manual work.*
- *To provide quick access to attendance reports.*

4. PROJECT SCOPE

Features Included

- *Student attendance marking (Present/Absent).*
- *Batch-wise attendance management.*
- *Date-wise attendance filtering.*
- *Attendance percentage calculation.*
- *Dashboard view for attendance records.*

Future Enhancements

- *User authentication for trainers and administrators.*
- *Export attendance reports to PDF or Excel.*
- *Email or SMS notifications for absent students.*
- *Database integration and cloud storage.*

5. TECHNOLOGIES USED

- *React.js*
- *JavaScript (ES6)*
- *HTML5*
- *CSS3*
- *Bootstrap*
- *React Hooks (useState, useEffect)*
- *Local Storage (for storing attendance data)*

6. APPLICATION FLOW

The trainer opens the dashboard. A batch is selected from the available batches. The current date is displayed automatically. Attendance is marked for each student as Present or Absent. Attendance records are saved in local storage. Trainers can view attendance records batch-wise. Date filters can be applied to view attendance for a specific day. The system calculates and displays attendance percentages automatically.

7. KEY REACT CONCEPTS USED

Components

Used to create reusable UI sections such as Dashboard, Attendance Form, and Attendance List.

useState Hook

Used for managing attendance data, selected batch, and date filters.

useEffect Hook

Used to load saved attendance data and update records when changes occur.

Props

Used to pass data between parent and child components.

Event Handling

Implemented for button clicks and form submissions.

Conditional Rendering

Used to display attendance percentages and filtered records dynamically.

Local Storage

Used for storing attendance records without requiring a backend database.

8. EXPECTED OUTPUT

Trainer Session Attendance

Total Students
6

Present
3

Absent
3

All

Present

Absent

ID	Name	Session	Status
1	Ar Jun	React Basics	Present
2	Priya	React Basics	Absent
3	Karthik	API Handling	Present
4	Meena	API Handling	Present
5	Ravi	React Basics	Absent
6	Sneha	API Handling	Absent

Trainer Session Attendance

Total Students
6

Present
3

Absent
3

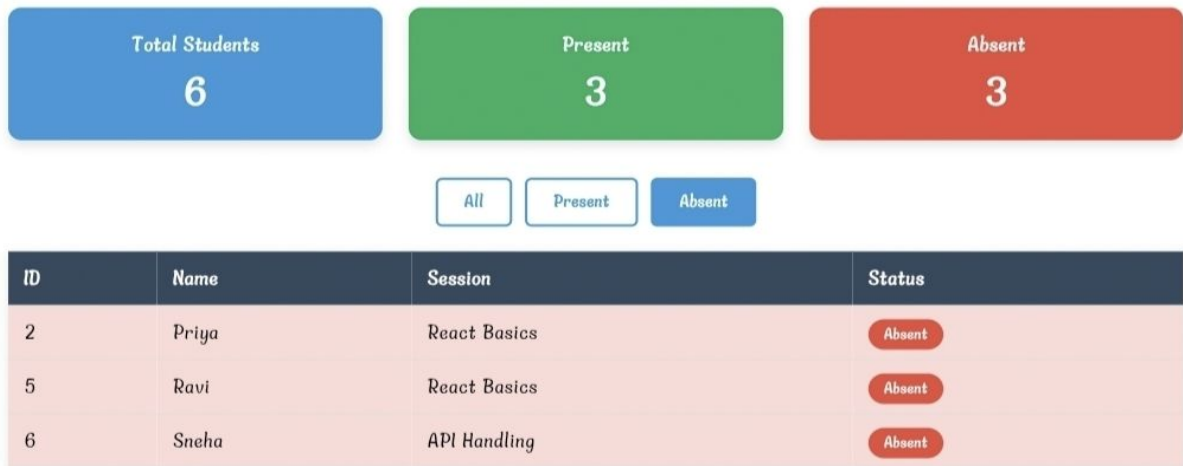
All

Present

Absent

ID	Name	Session	Status
1	Ar Jun	React Basics	Present
3	Karthik	API Handling	Present
4	Meena	API Handling	Present

Trainer Session Attendance



9. CHALLENGES FACED

- Managing state updates efficiently using React Hooks.
- Implementing date-wise filtering accurately.
- Calculating attendance percentages dynamically.
- Ensuring proper storage and retrieval of attendance records from local storage.

These challenges were solved by using React Hooks effectively and organizing components properly.

10. LEARNING OUTCOMES

- Understanding React component architecture.
- Working with `useState` and `useEffect` hooks.
- Managing and storing data using Local Storage.
- Implementing filtering and dynamic rendering.
- Designing responsive user interfaces.
- Improving problem-solving and debugging skills.

11. CONCLUSION

The Trainer Session Attendance Dashboard successfully provides an efficient solution for managing student attendance across multiple batches. The application simplifies attendance recording, supports batch-wise and date-wise viewing, and automatically calculates attendance percentages. The project enhanced practical knowledge of React and demonstrated how modern web technologies can be used to solve real-world problems effectively.